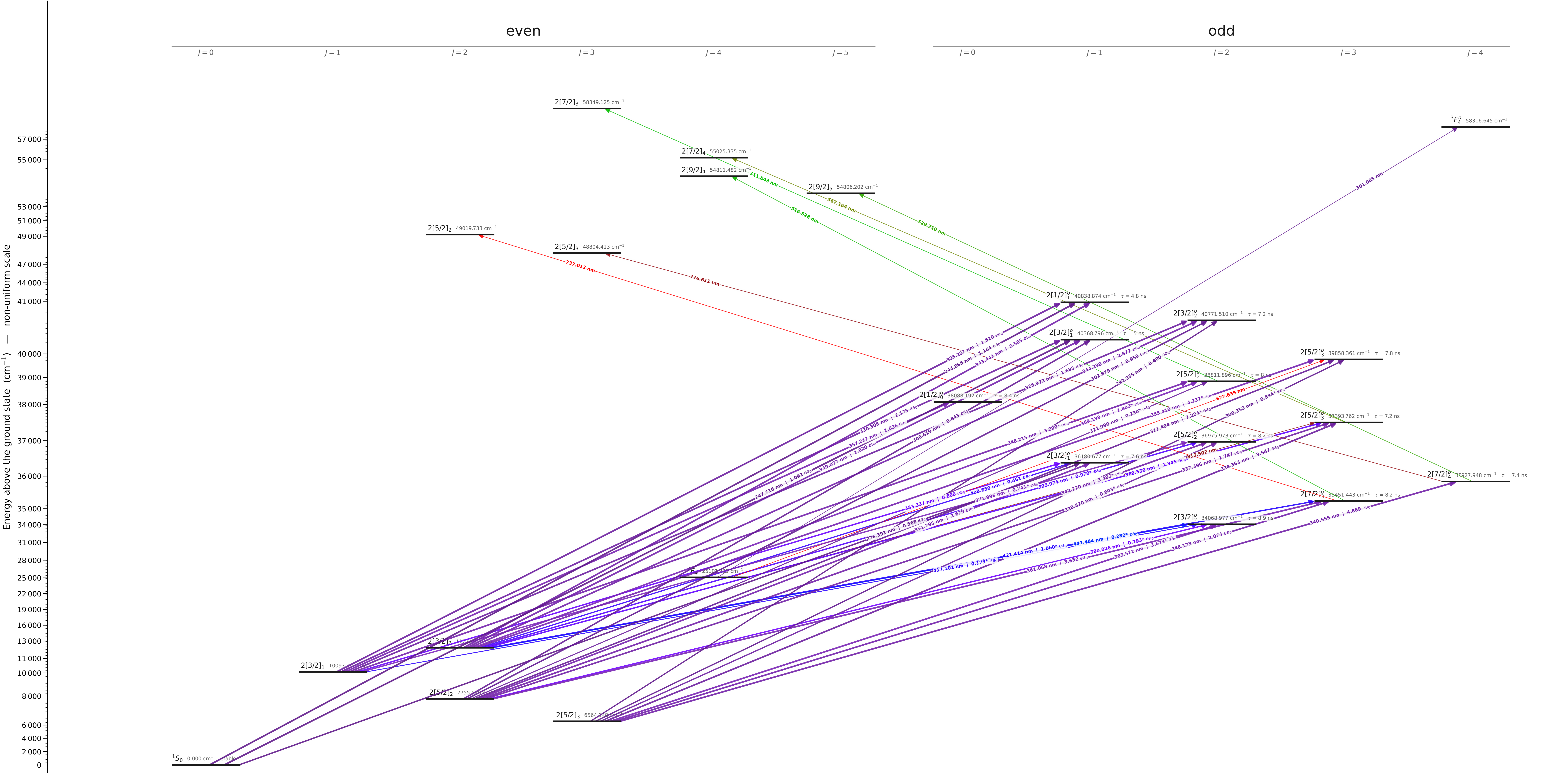
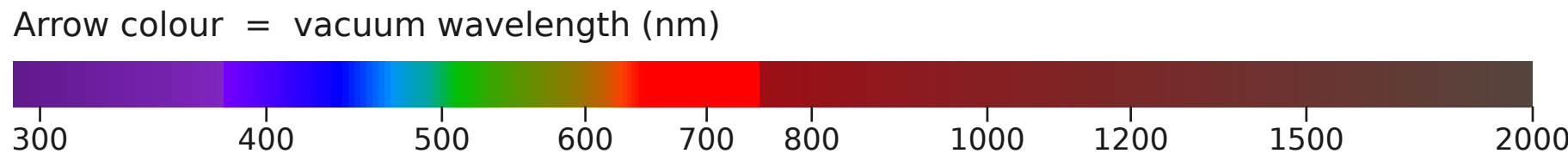


Palladium

Pd (Pd I). Strongest classified transitions below 2 μm (NIST): 25 levels, 46 lines.

Pd⁺ ionisation limit 67241.14 cm⁻¹ = 8.33684 eV (148.718 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle U || e r || U' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er U'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ - 2[1/2] ₀ ^o	244.8647	244.7906	1224.3186	1.164	6.233e+07	33.16	28.6	ground-state resonance line 244.9 nm
¹ S ₀ - 2[3/2] ₀ ^o	247.7161	247.6413	1210.2261	1.092	5.300e+07	31.83	26.5	ground-state resonance line 247.7 nm
¹ S ₀ - 2[3/2] ₁ ^o	276.3906	276.3090	1084.6694	0.568	1.033e+07	20.94	7.9	ground-state resonance
2[5/2] ₃ - 2[7/2] ₃ ^o	340.5554	340.4577	880.3046	4.869	1.351e+08	21.51	100	closed j=3 -> j=4 cycling transition from the metastable...
2[3/2] ₁ - 2[1/2] ₀ ^o	357.2169	357.1149	839.2450	1.636	1.190e+08	18.95	100	357.2 nm line used for collinear laser spectroscopy tests
2[5/2] ₂ - 2[7/2] ₂ ^o	361.0575	360.9546	830.3177	3.652	8.200e+07	19.41	67.2	361.1 nm
2[5/2] ₃ - 2[3/2] ₂ ^o	363.5725	363.4689	824.5740	3.673*	1.137e+08	17.88	100	363.5 nm line from metastable 3D3: main line for collinea...
2[3/2] ₂ - 2[5/2] ₂ ^o	369.1387	369.0337	812.1404	1.803*	2.618e+07	19.89	20.9	369.1 nm line used for collinear laser spectroscopy tests

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁰⁵Pd hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[5/2] ₃	-391.1790(10)	-652.906(15)	F=1/2: +3520.05 F=3/2: +3109.56 F=5/2: +2360.13 F=7/2: +1173.82 F=9/2: -586.49 F=11/2: -3097.07
2[5/2] ₂	66.3590(10)	-398.192(10)	F=1/2: -743.25 F=3/2: -464.52 F=5/2: -99.53 F=7/2: +202.41 F=9/2: +232.25
2[3/2] ₂	-83.5(3)	-304(3)	F=1/2: +371.63 F=3/2: +383.23 F=5/2: +326.52 F=7/2: +87.49 F=9/2: -493.52
2[7/2] ₃ ^o	-115.3(3)	-497(4)	F=1/2: +854.80 F=3/2: +816.04 F=5/2: +701.74 F=7/2: +437.35 F=9/2: -81.50 F=11/2: -989.00
2[7/2] ₂ ^o	-79.7(3)	-469.9(18)	F=3/2: +765.50 F=5/2: +704.70 F=7/2: +566.72 F=9/2: +298.69 F=11/2: -167.35 F=13/2: -914.48
2[3/2] ₁ ^o	-126.2(6)	6.0(20)	F=3/2: +445.90 F=5/2: +121.40 F=7/2: -314.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
¹ S ₀ - 2[3/2] ₀ ^o	276.391	105-108	77(3)
2[5/2] ₃ - 2[7/2] ₃ ^o	340.555	104-106	468.6(27)
2[5/2] ₃ - 2[7/2] ₂ ^o	340.555	105-106	347(3)
2[5/2] ₃ - 2[7/2] ₁ ^o	340.555	108-106	-505(4)
2[5/2] ₃ - 2[7/2] ₀ ^o	340.555	110-106	-947(5)
2[5/2] ₂ - 2[7/2] ₃ ^o	340.555	102-106	938(18)
2[3/2] ₁ - 2[1/2] ₀ ^o	357.217	102-108	1446(4)
2[3/2] ₁ - 2[1/2] ₁ ^o	357.217	104-108	958(3)
2[3/2] ₁ - 2[1/2] ₂ ^o	357.217	105-108	842.0(20)
2[3/2] ₁ - 2[1/2] ₃ ^o	357.217	106-108	490.0(20)
2[3/2] ₁ - 2[1/2] ₄ ^o	357.217	110-108	-443(5)
2[5/2] ₂ - 2[7/2] ₂ ^o	361.058	102-108	1413.6(14)
2[5/2] ₂ - 2[7/2] ₁ ^o	361.058	104-108	948.4(11)
2[5/2] ₂ - 2[7/2] ₀ ^o	361.058	105-108	844.7(9)
2[5/2] ₂ - 2[7/2] ₃ ^o	361.058	106-108	486.2(11)
2[5/2] ₂ - 2[7/2] ₄ ^o	361.058	110-108	-429.4(15)
2[5/2] ₂ - 2[3/2] ₂ ^o	363.572	104-106	467.1(14)
2[5/2] ₂ - 2[3/2] ₁ ^o	363.572	105-106	348.4(16)
2[5/2] ₂ - 2[3/2] ₀ ^o	363.572	108-106	-497.2(15)
2[5/2] ₂ - 2[3/2] ₃ ^o	363.572	110-106	-939.2(27)
2[3/2] ₂ - 2[5/2] ₂ ^o	369.139	102-108	1337(4)
2[3/2] ₂ - 2[5/2] ₁ ^o	369.139	104-108	889(4)

Shift = ν(first isotope) - ν(second isotope).